London Ambulance Service
NHS Trust

Patient Assessment

Lead II Electrocardiography (ECG)

Lead II ECG

The ECG is only an investigation and **must** always be interpreted in light of the **clinical** findings.

Severe cardiac disease may be present despite a normal ECG

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Lead II ECG

- History and clinical examination is very important
- Lead II is inadequate for diagnosis
- 12 Lead ECG is essential to make a diagnosis
- Attach a paper copy ECG with the PRF
- Photograph ECG for ePCR

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Lead II ECG

ECG monitoring - lead position

Four Electrode Positioning (Limbs)

- **Ride** RA
- **Your** LA
- **Green** LL
- **Bike** RL



Tips…

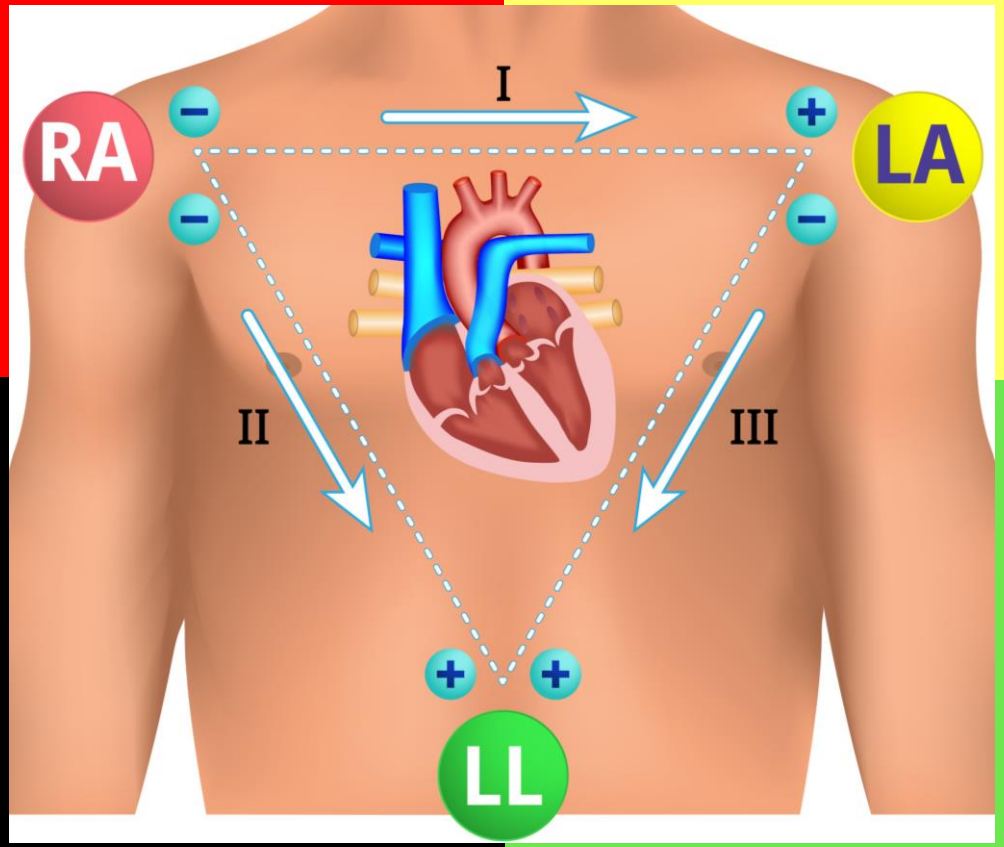
- ✓ Avoid crossed wires
- ✓ Equidistant leads
  - ✓ In-date sealed pack dots
- ✓ Prepare the skin
  - ✓ Dry
  - ✓ Free from loose skin
  - ✓ Avoid hairy areas

It's ideal to obtain an ECG on a trolley or bed with the patient semi-recumbent

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Ride

Your

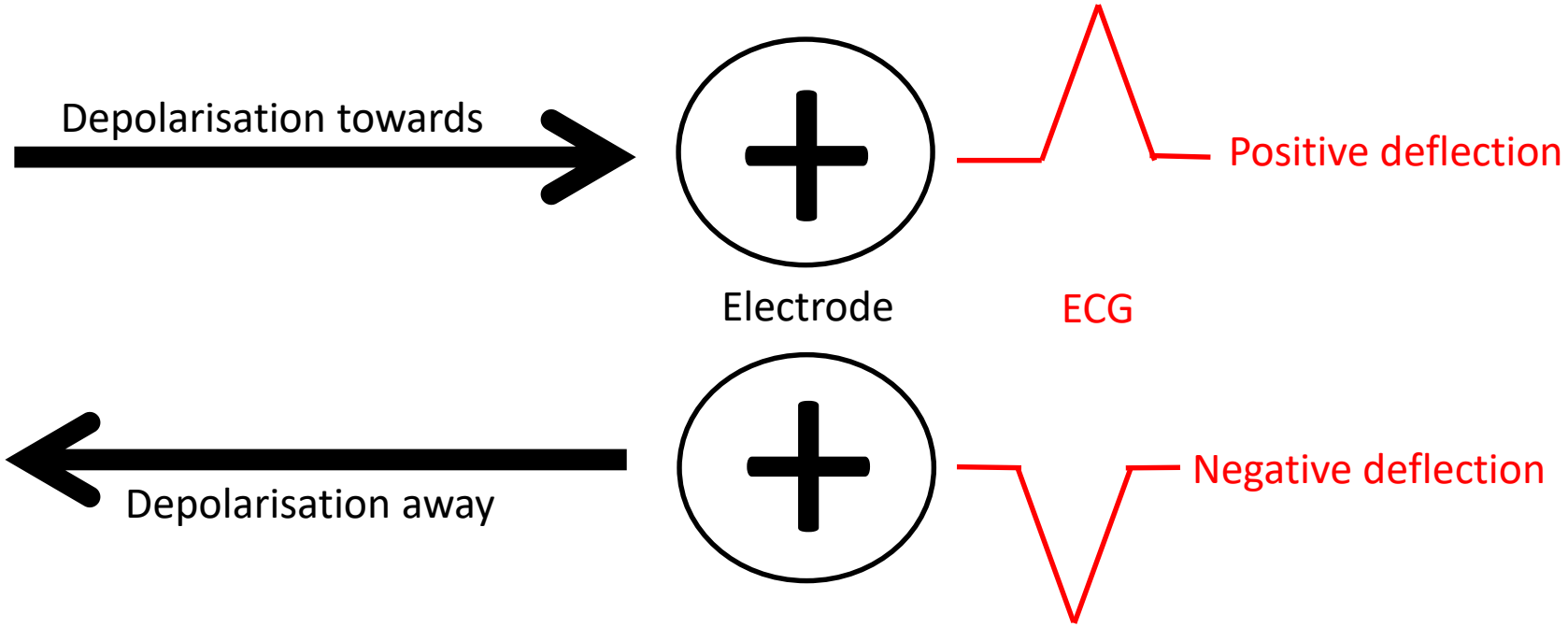


Bike

Green

Lead II ECG

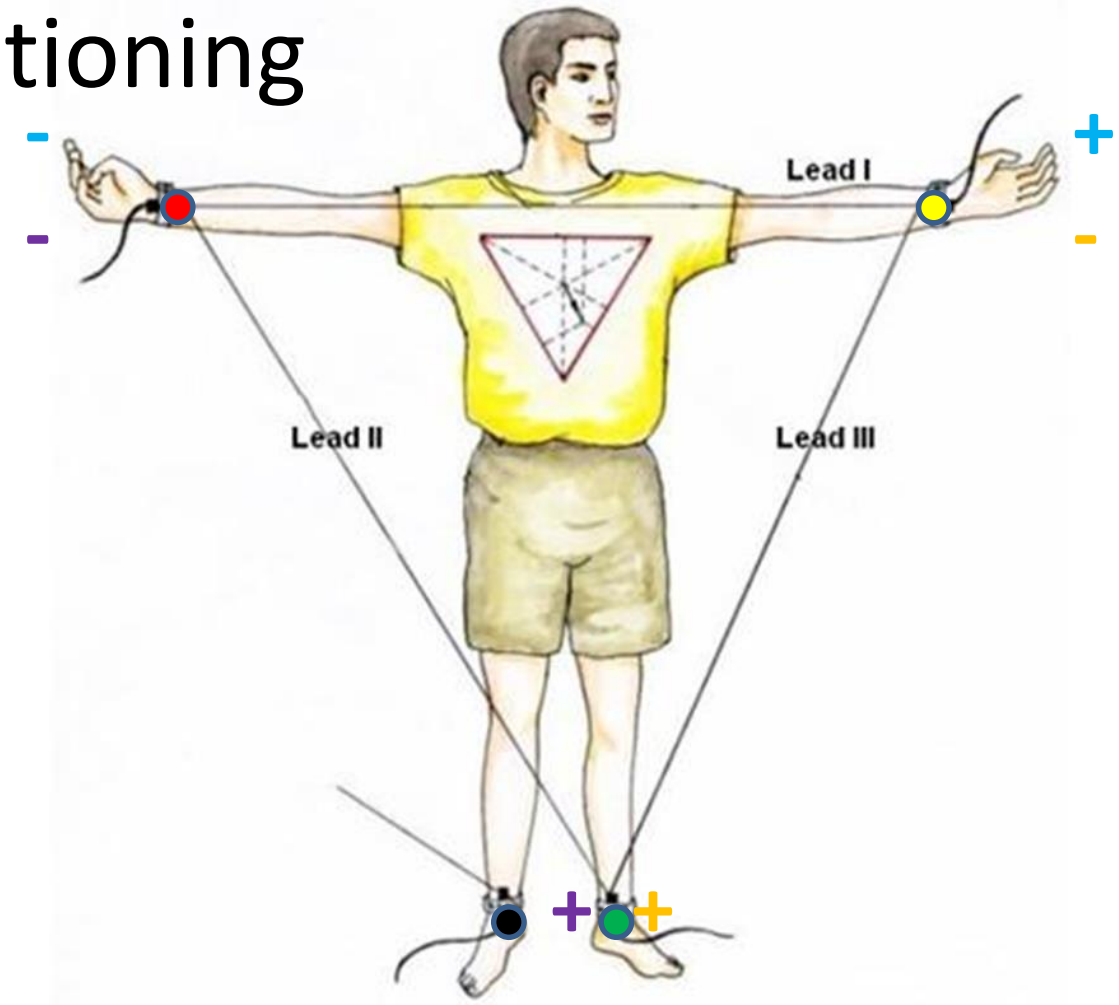
Depolarisation



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Lead II ECG

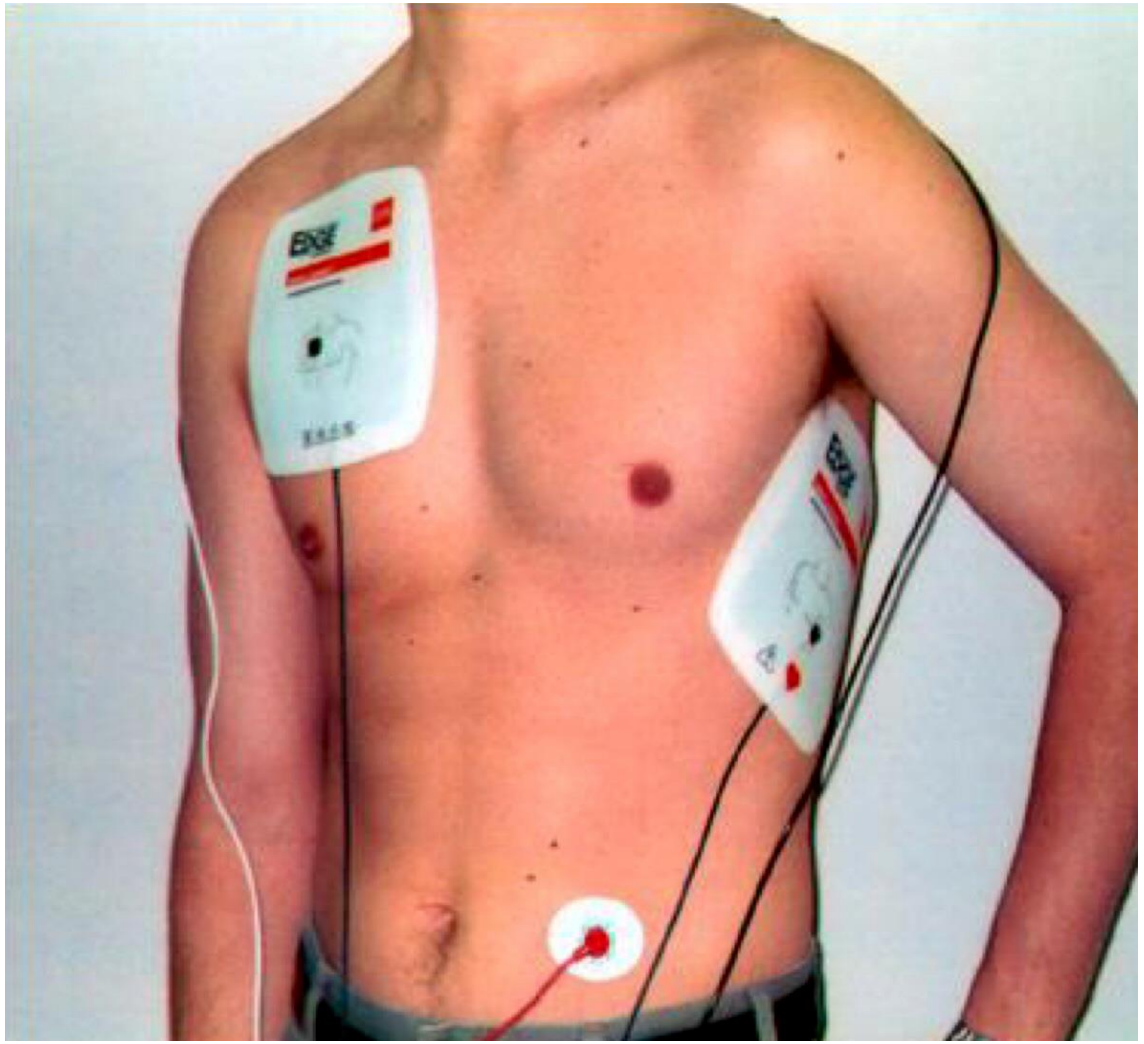
Lead positioning



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Lead II ECG

Defibrillator Pad Placement



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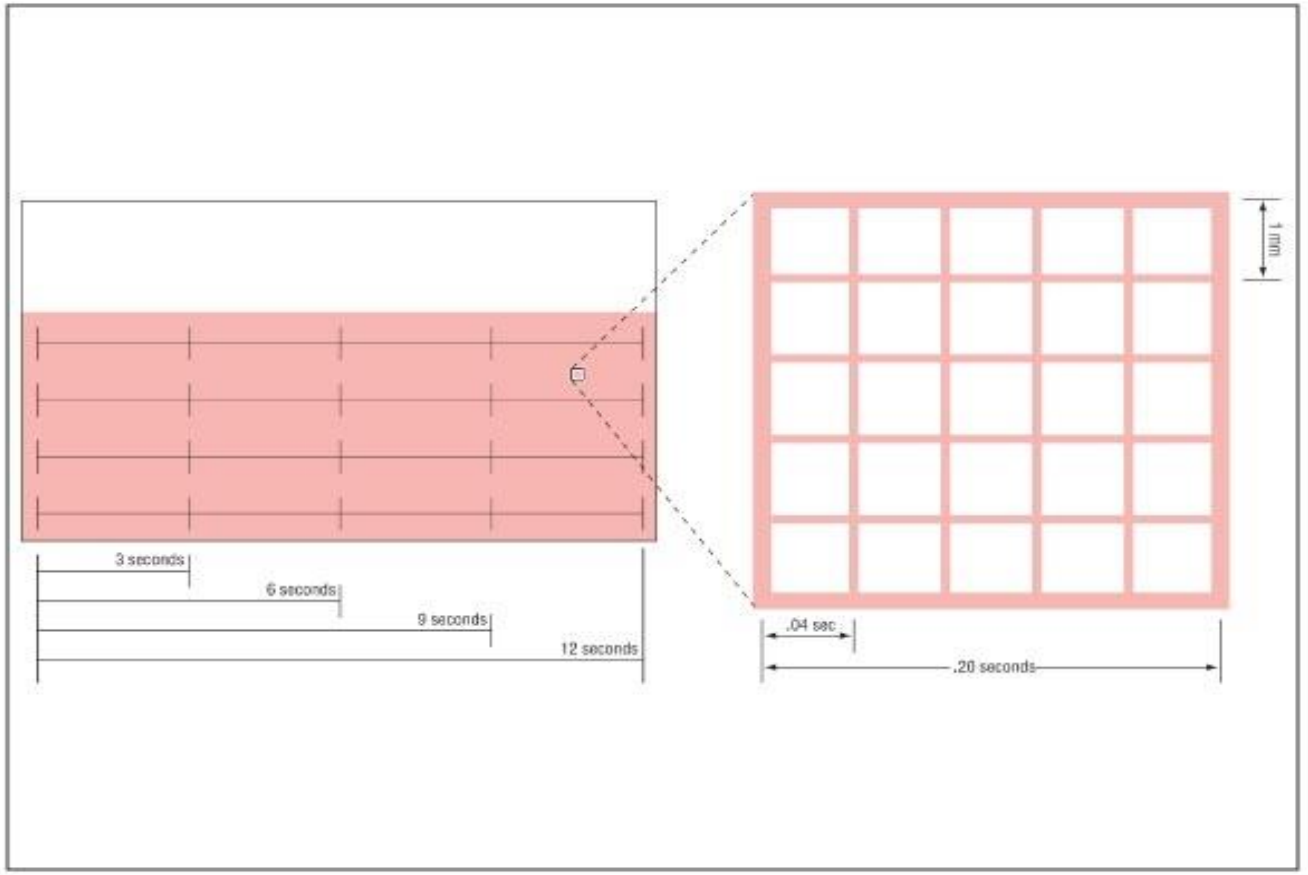
Lead II ECG

ECG paper

ECG paper. Height (amplitude) is measured in millimetres (mm) and width in milliseconds (ms).

Figure 27.24

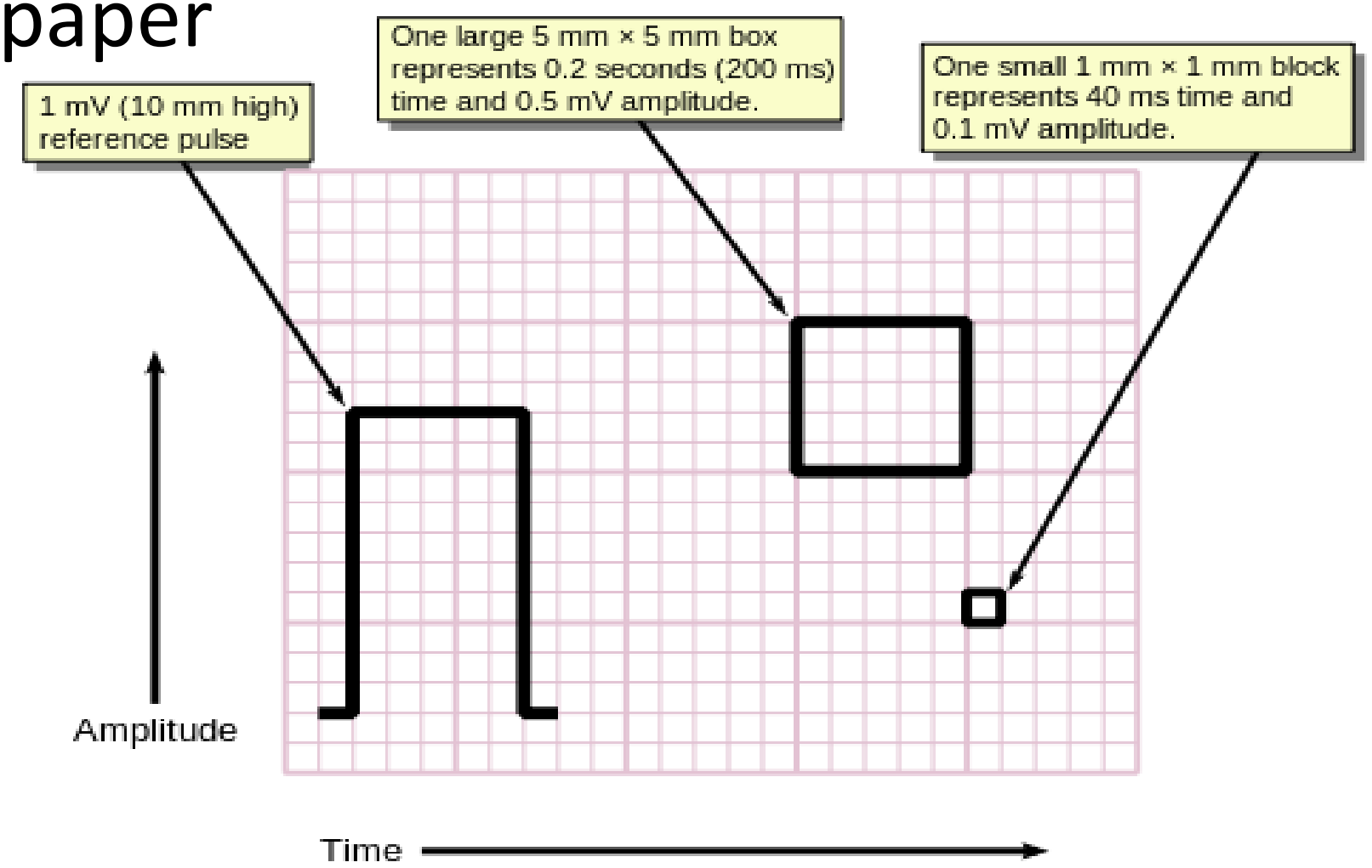
From *12-Lead ECG: The Art of Interpretation*, courtesy of Tomas B. Garcia, MD.



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Lead II ECG

ECG paper



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Lead II ECG

Normal Values

Each SMALL square = 40 Msec (0.04 sec)

Each LARGE square = 200 Msec (5 x 40)

Normal PR interval (adults) 120 - 200 Msec (3-5 small boxes)

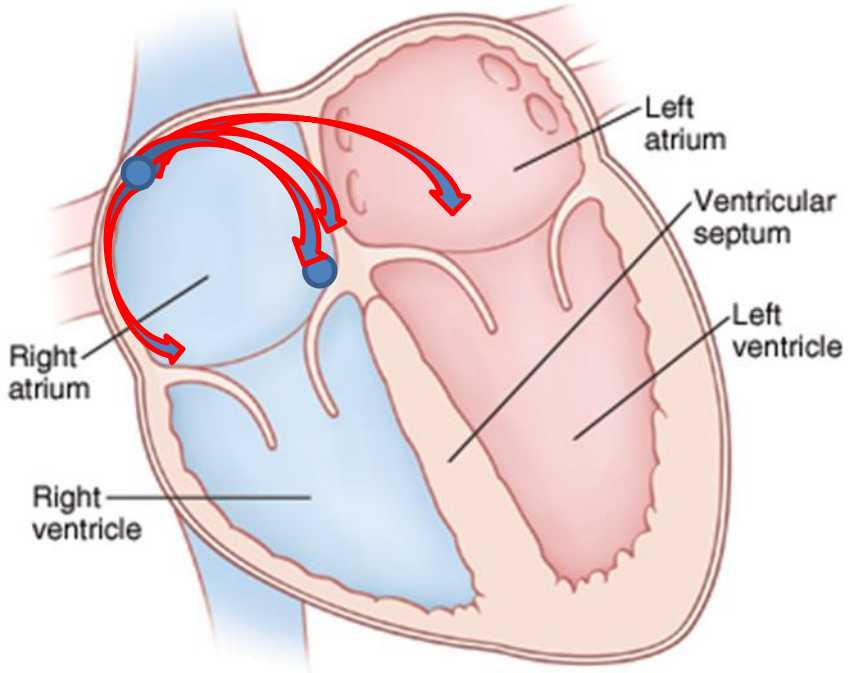
Normal QRS interval (adults) 80 - 120 Msec (2-3 small boxes)

QTc Interval < 500 Msec

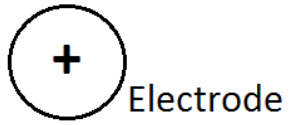
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Lead II ECG

P – wave Depolarisation of Atria





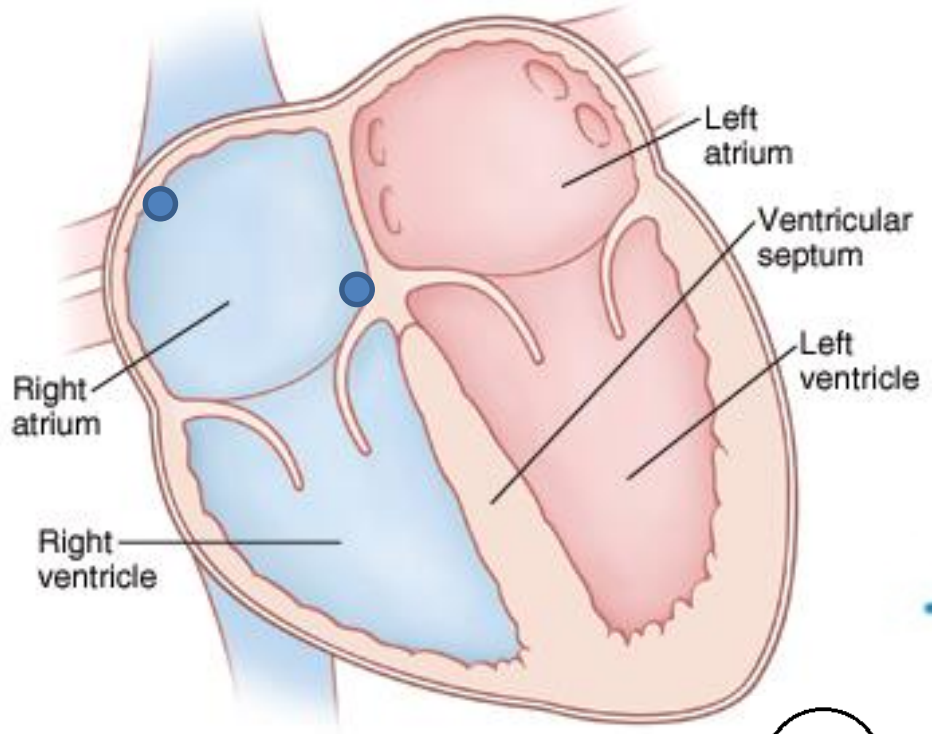


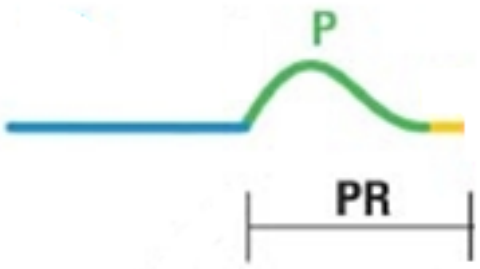
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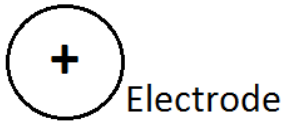
Lead II ECG

P-R interval

Delay at AV junction





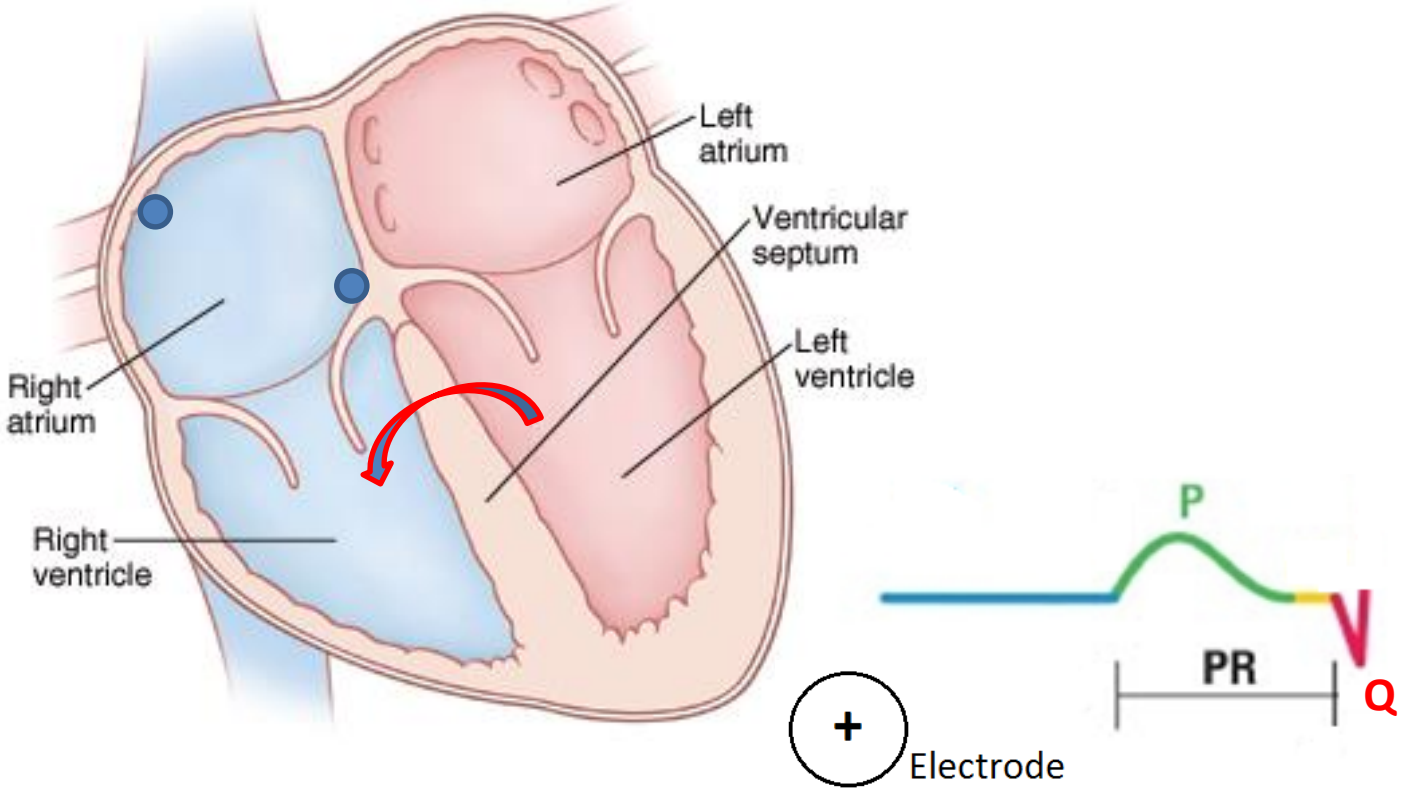


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Lead II ECG

Q – wave

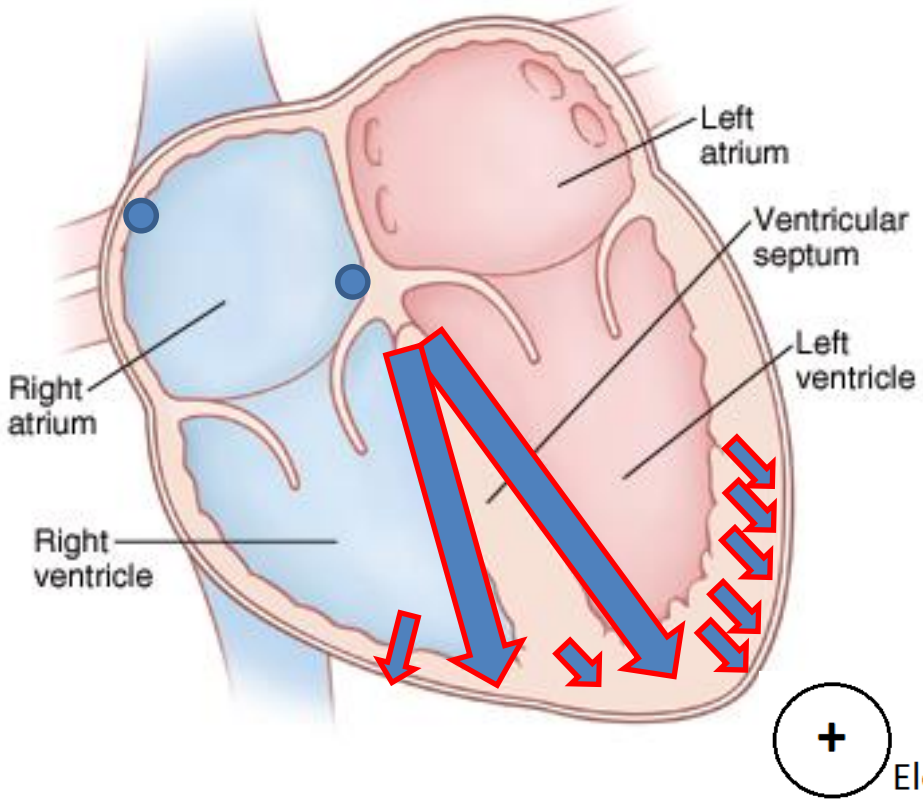
Septal depolarisation

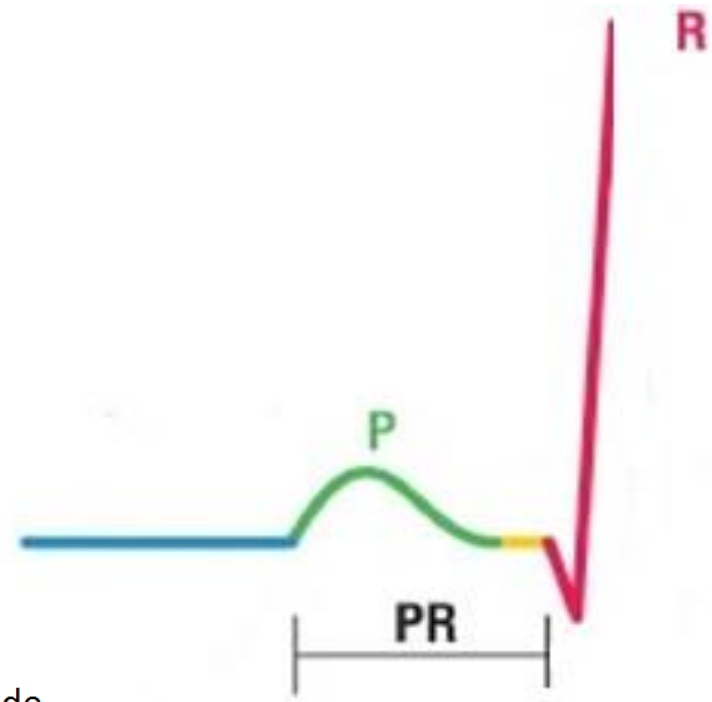


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Lead II ECG

R – wave Ventricular depolarisation



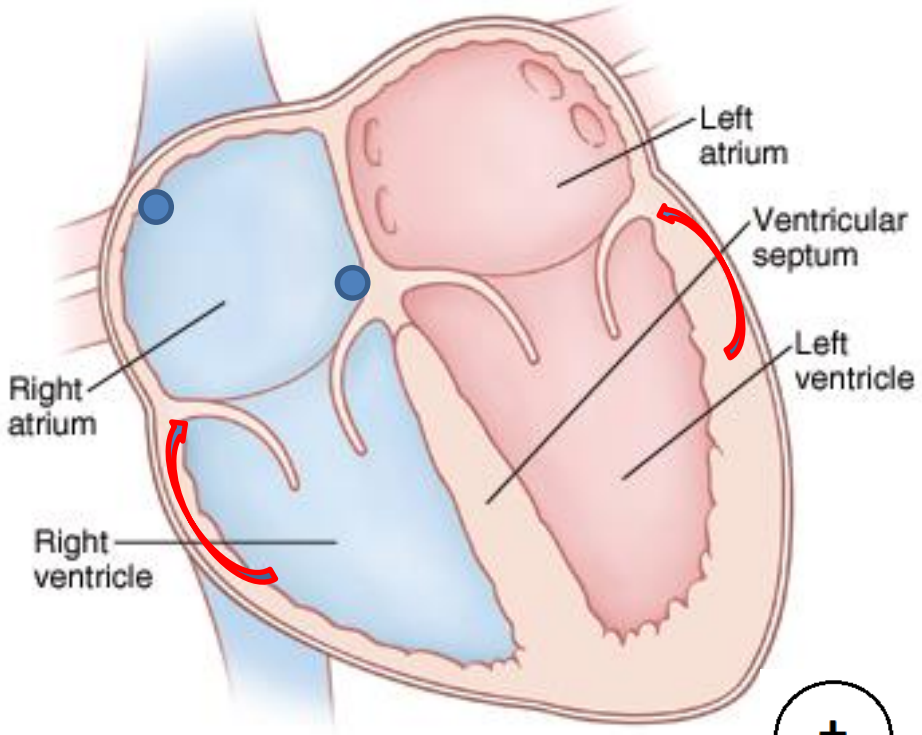


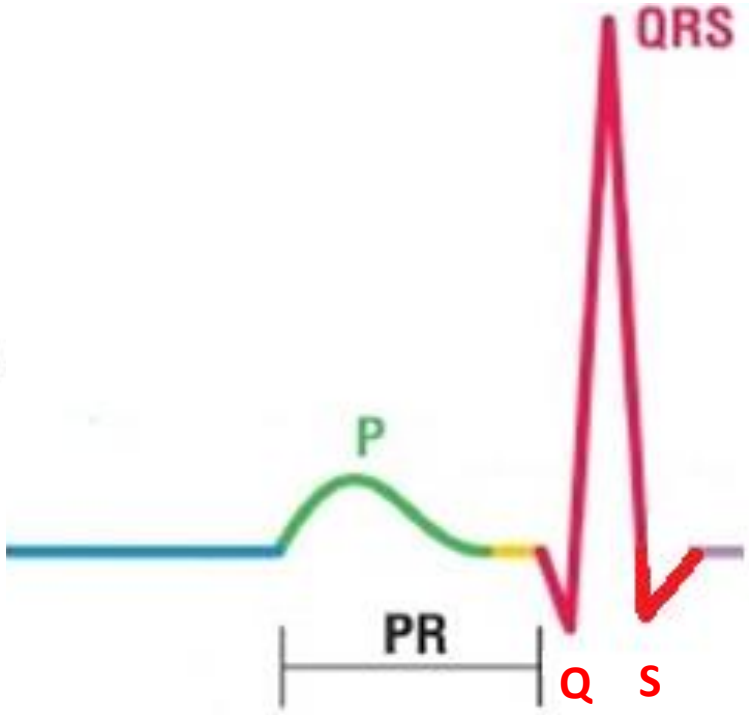
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Lead II ECG

S – wave

Late stage depolarisation



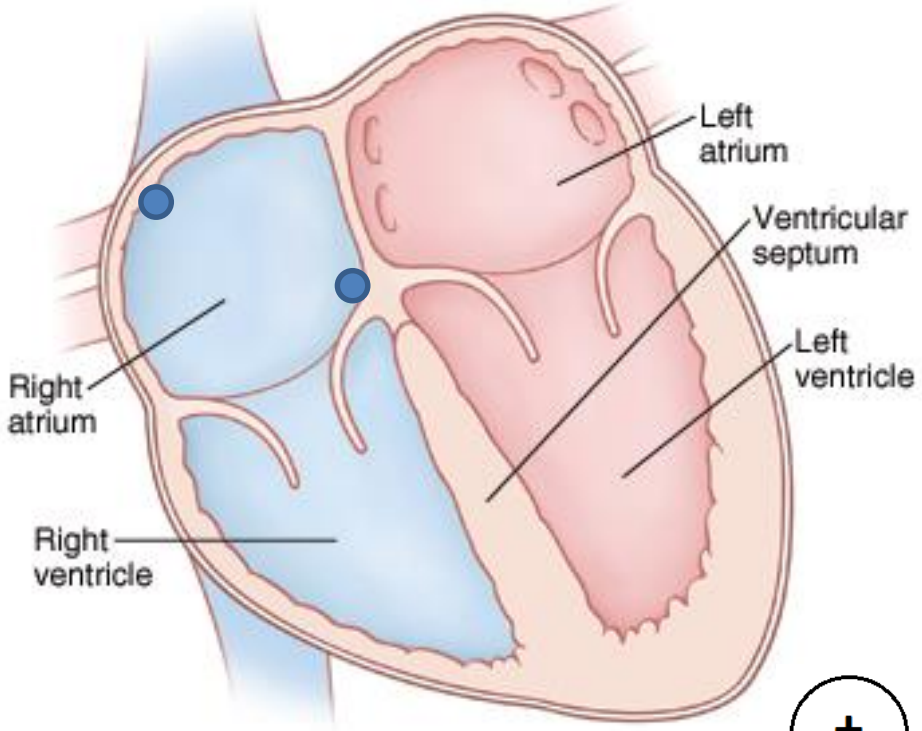


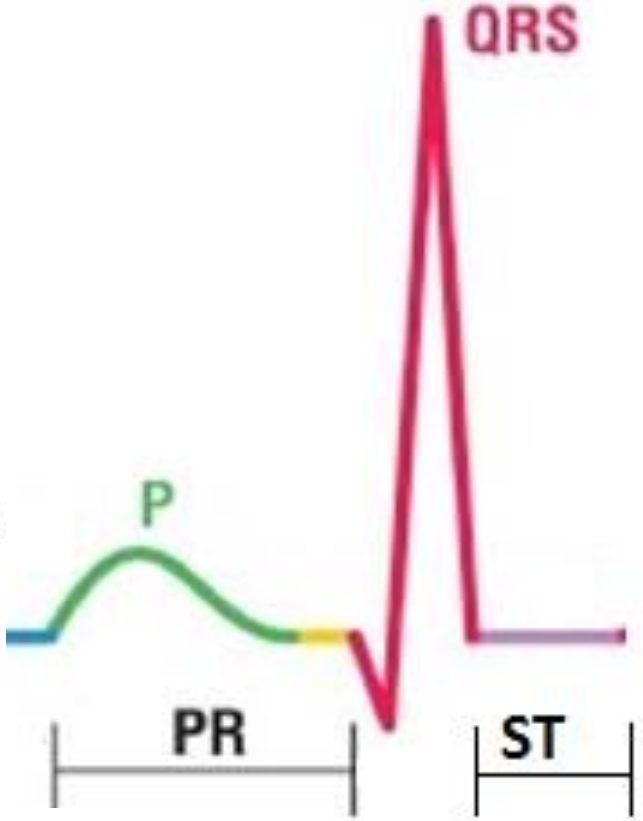
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Lead II ECG

S – T segment

Isoelectric

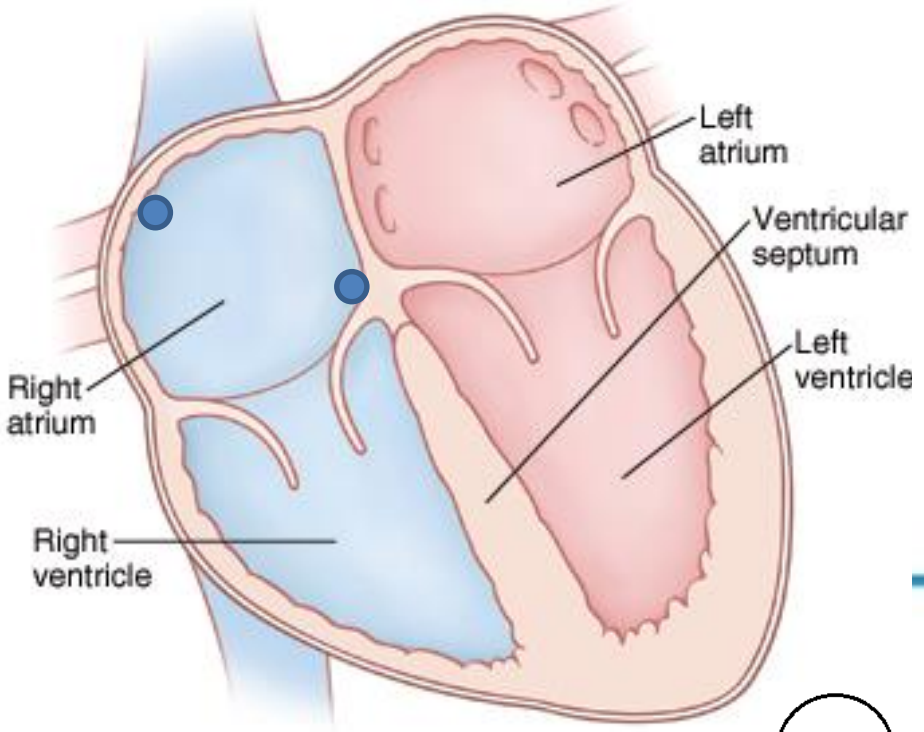


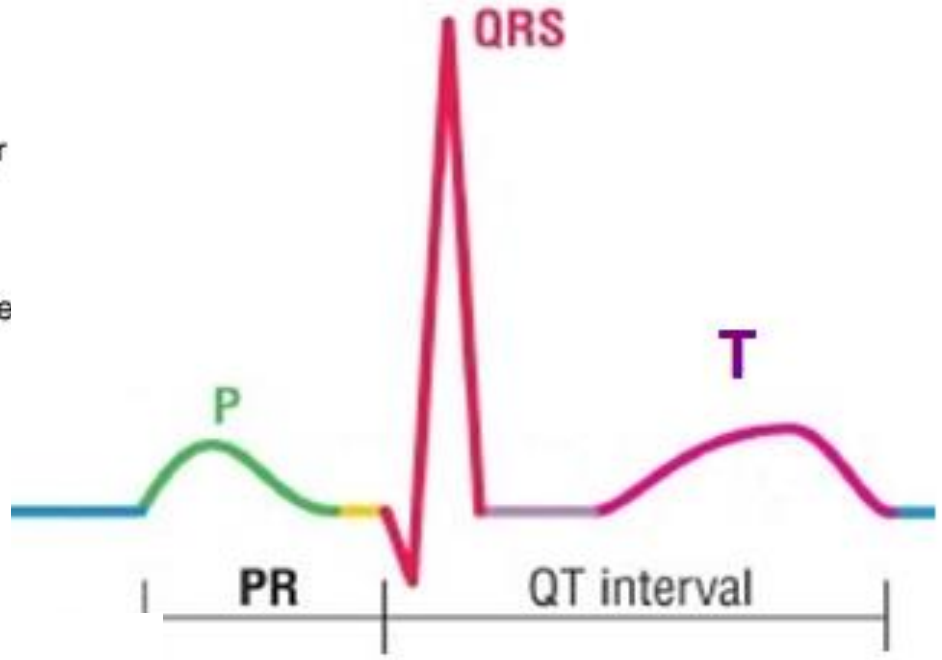


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Lead II ECG

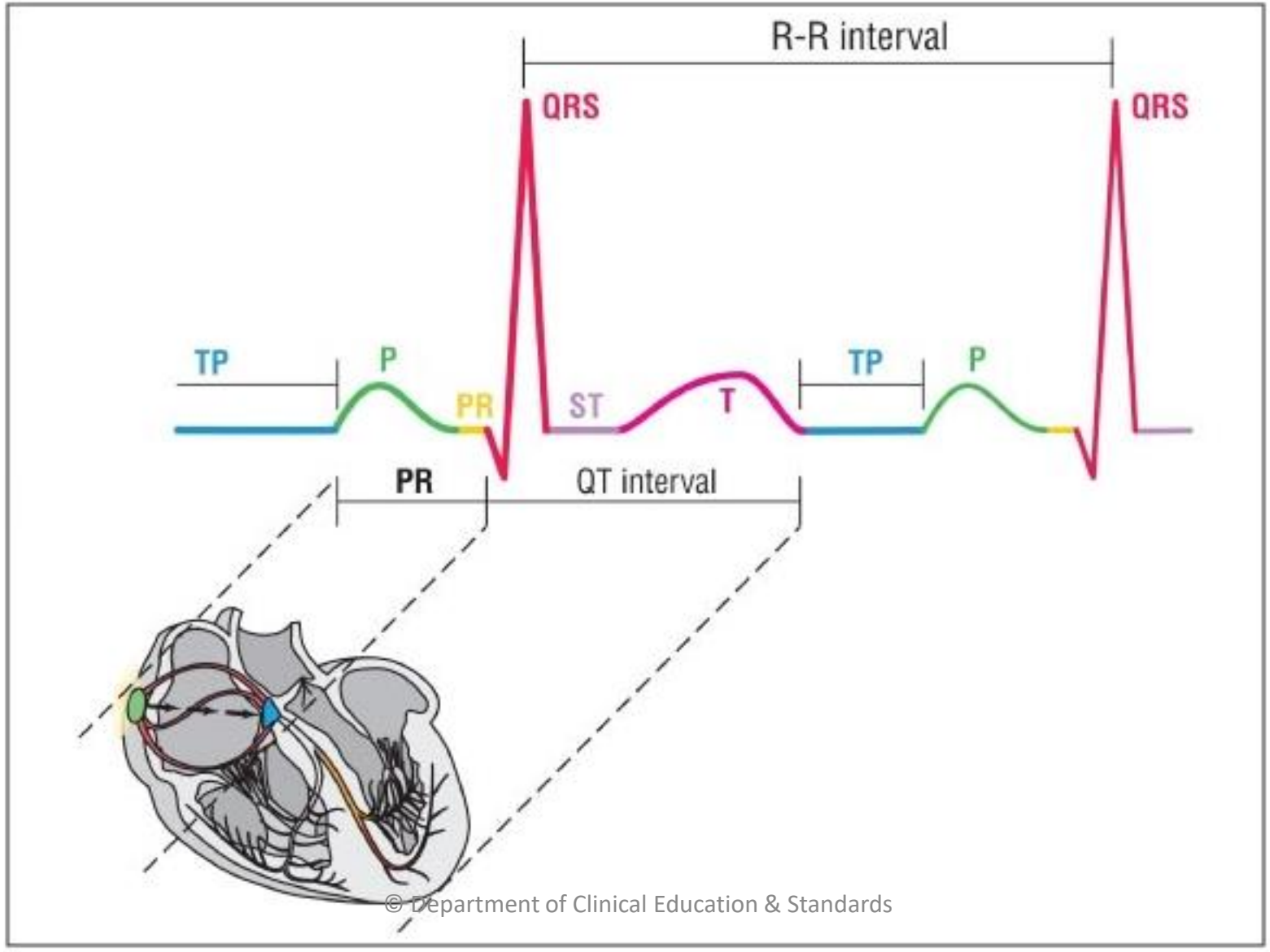
T – wave Repolarisation of ventricles





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Lead II ECG



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Lead II ECG

If the rhythm strip **does not** show NSR

It is an abnormal rhythm

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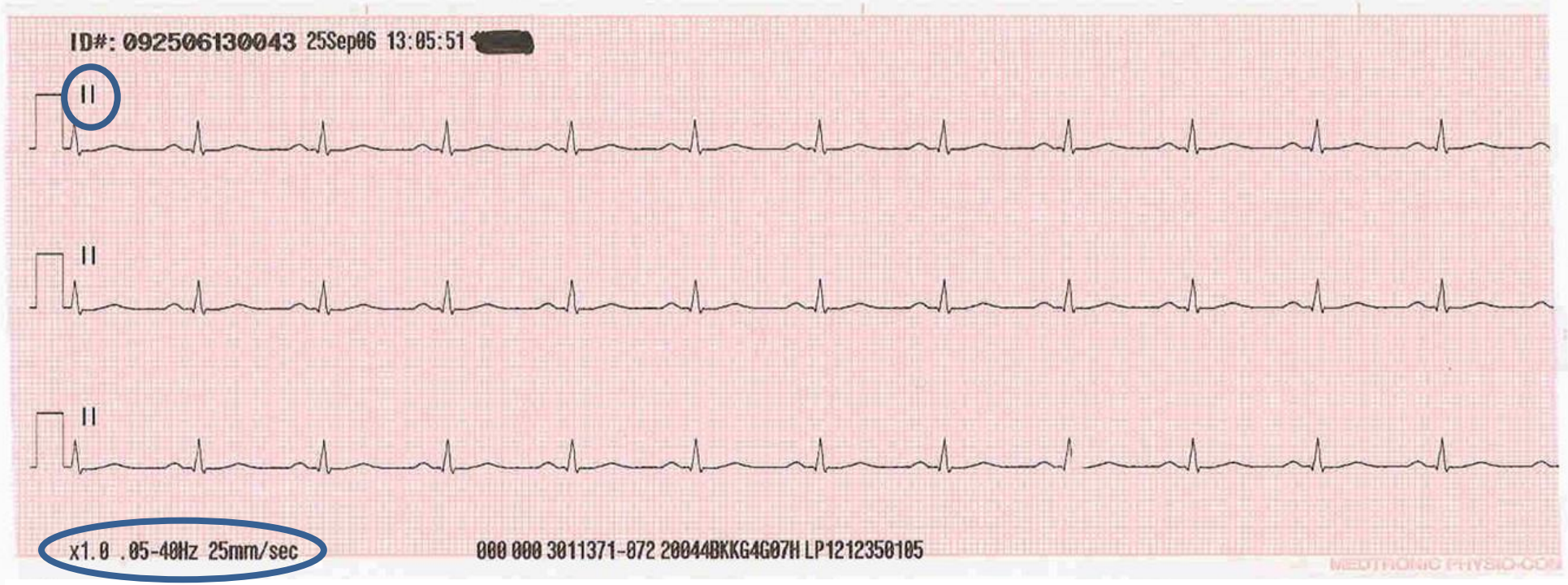
Lead II ECG

Lead II ECG - Rules for Interpretation

- Is there electrical activity?
- What is the rhythm?
- What is the rate?
- P Waves, shape, before every QRS, QRS after every P
- What is the P-R interval
- QRS complex, shape and duration

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Lead II ECG



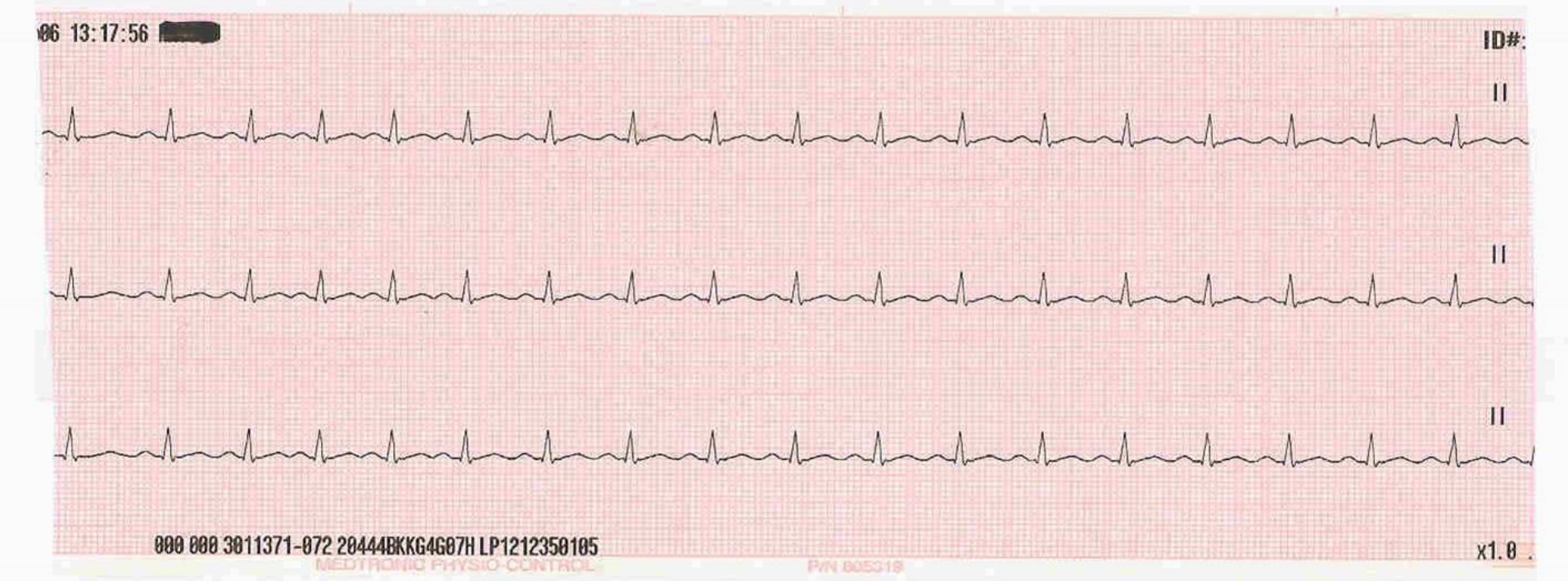
Rhythm **Normal Sinus Rhythm**

Electrical Activity? **YES** Pulse? **YES**

QRS Rate **80** Width **Narrow** Regular / Regularly Irregular / Irregular

Atria P – Waves / Atrial Activity? **P waves** Correlation **P waves before and after every QRS**

Lead II ECG



Rhythm

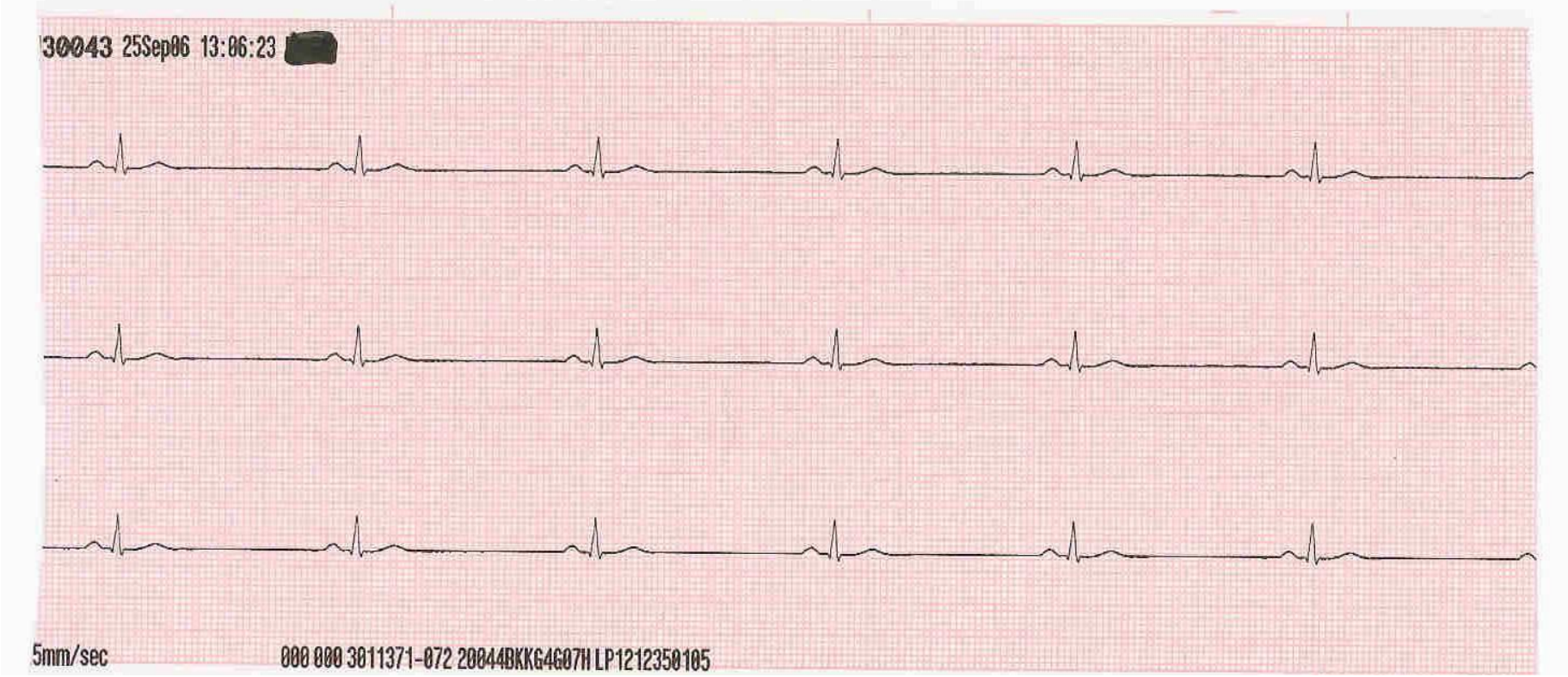
Electrical Activity? Pulse?

QRS Rate Width Regular / Regularly Irregular / Irregular

Atria P – Waves / Atrial Activity? Correlation

Valsalva manoeuvre Registered Paramedic skill only.

Lead II ECG



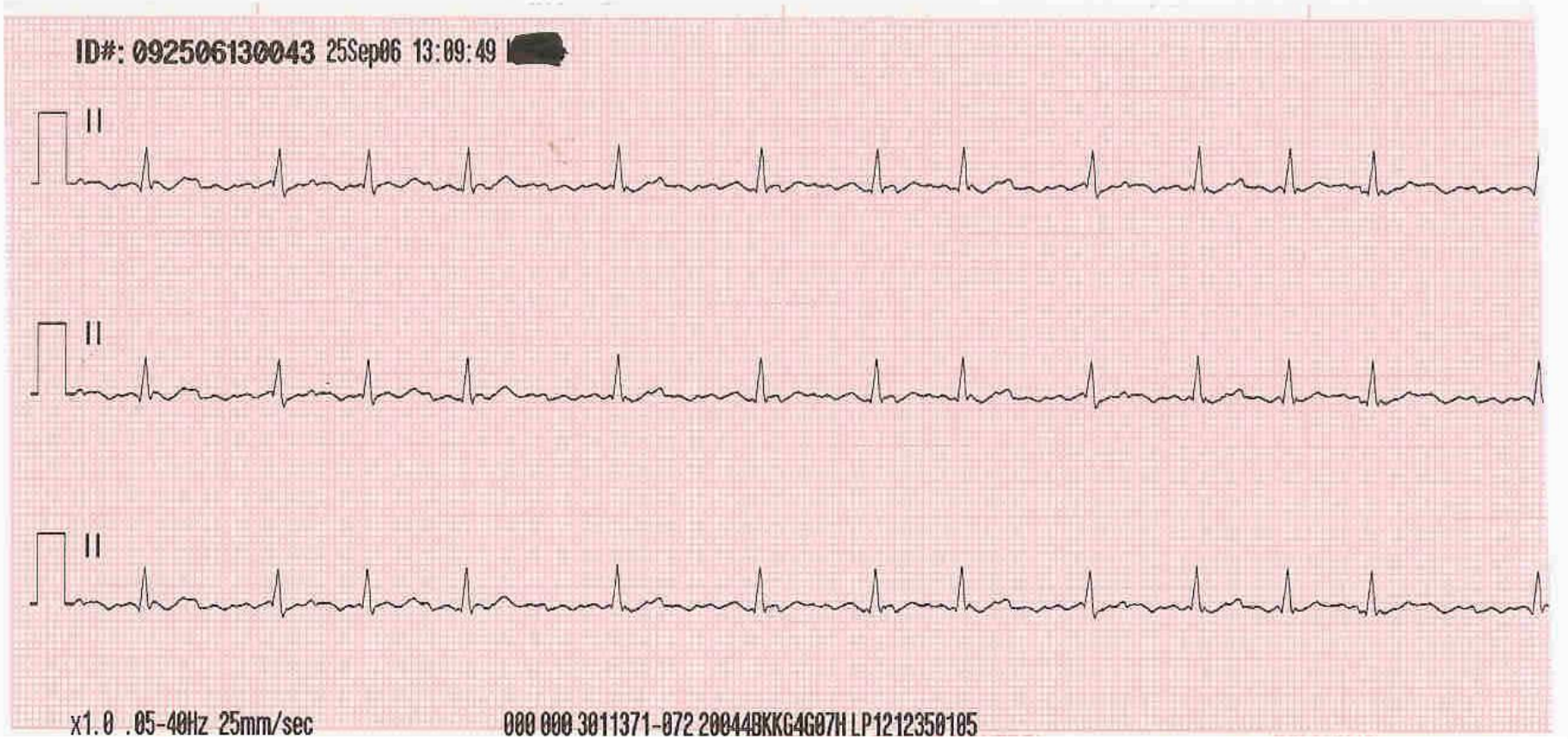
Rhythm

Electrical Activity? Pulse?

QRS Rate Width Regular / Regularly Irregular / Irregular

Atria P – Waves / Atrial Activity? Correlation

Lead II ECG



Rhythm

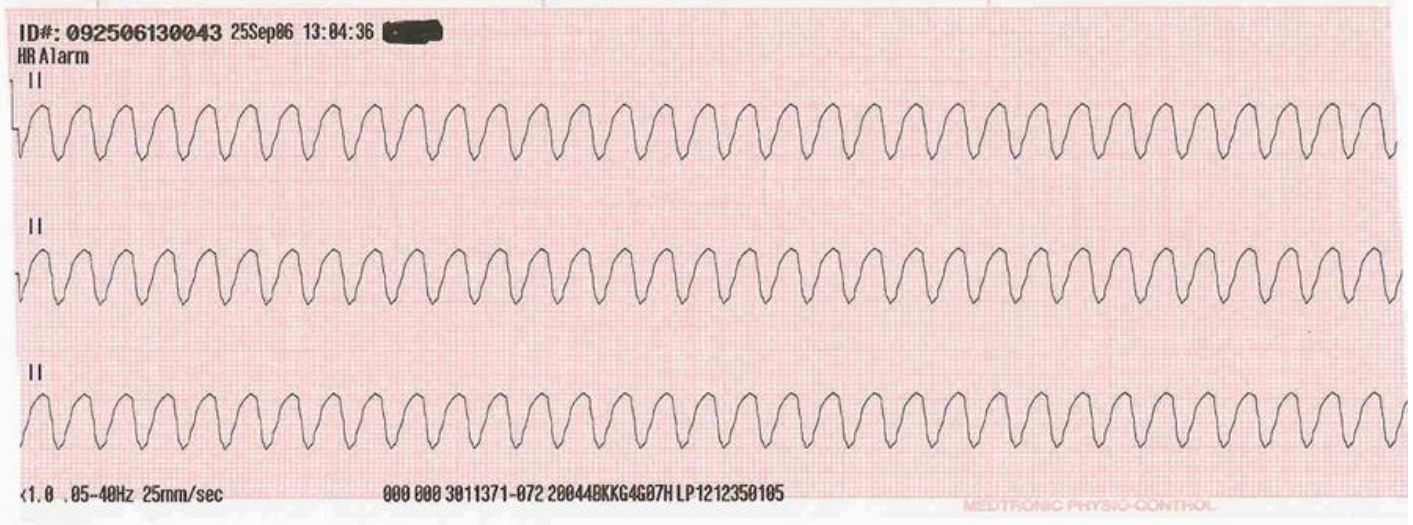
Electrical Activity? Pulse?

QRS Rate Width Regular / Regularly Irregular / Irregular

Atria P – Waves / Atrial Activity? Correlation

Lead II ECG

Ventricular Tachycardia (VT)



Regular

Rate >120 bpm

Difficult to identify any P waves

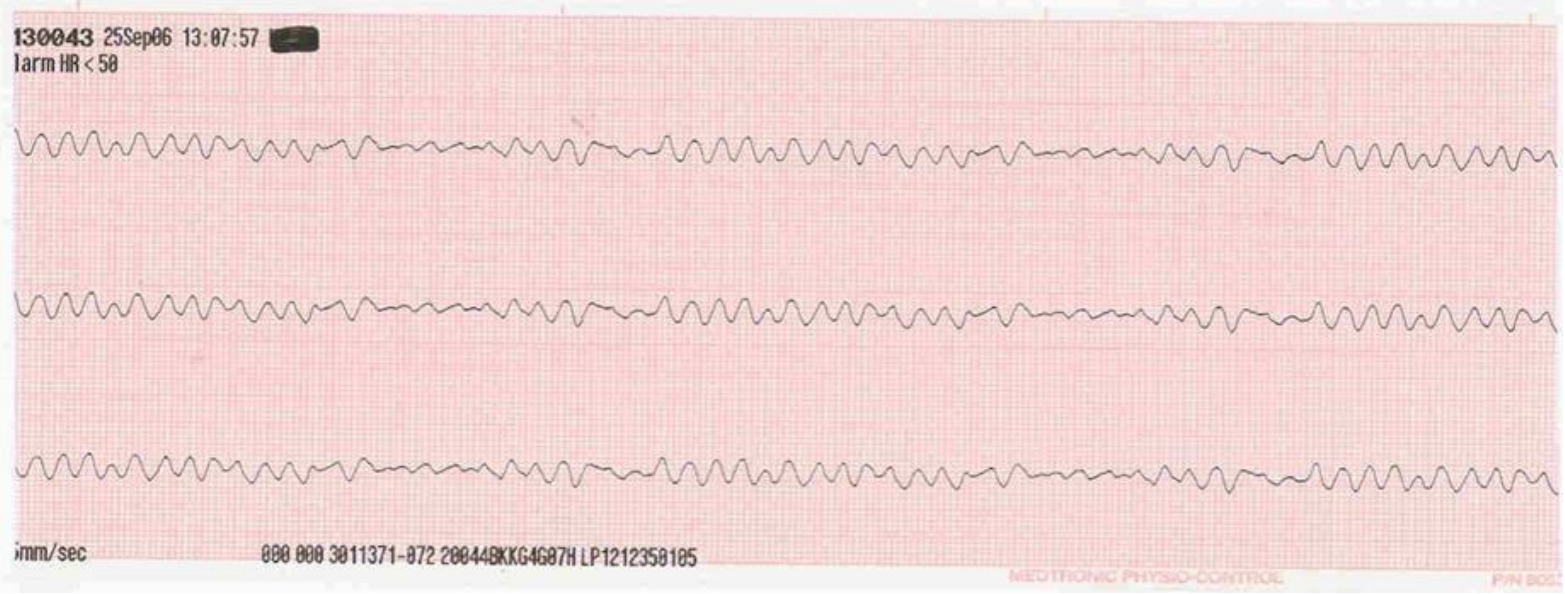
QRS wide *(more than 3 little boxes)* and bizarre looking

Pulsed or Pulseless?

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Lead II ECG

Ventricular Fibrillation (VF)

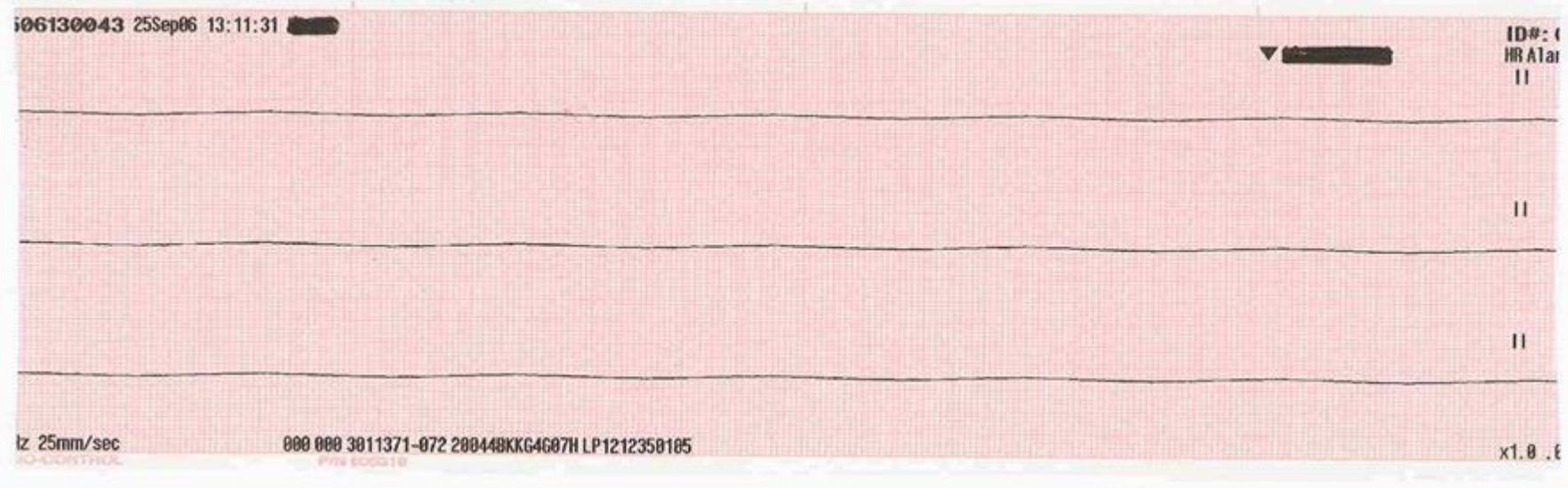


- **No pulse**
- Chaotic waveform
- Requires CPR and defibrillation

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Lead II ECG

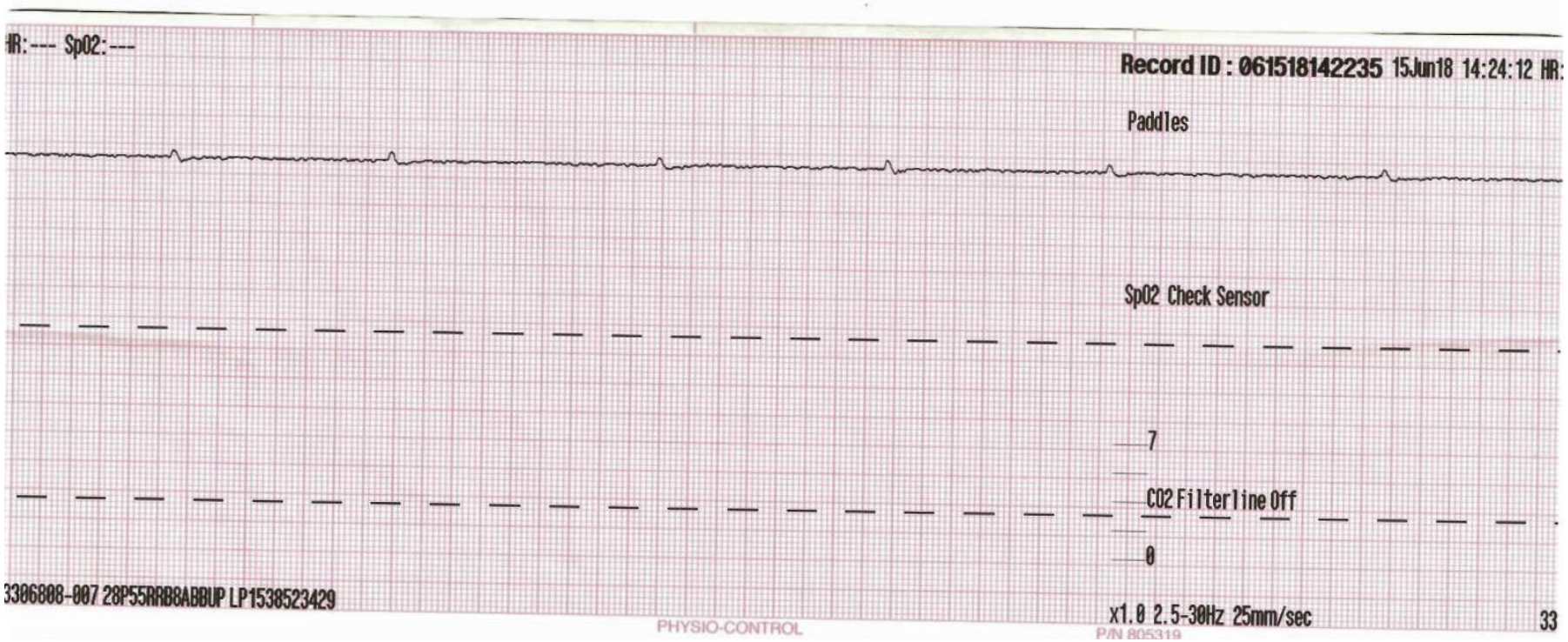
Asystole



- **No pulse**
- *Not a straight line*
- CPR

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Lead II ECG



Rhythm

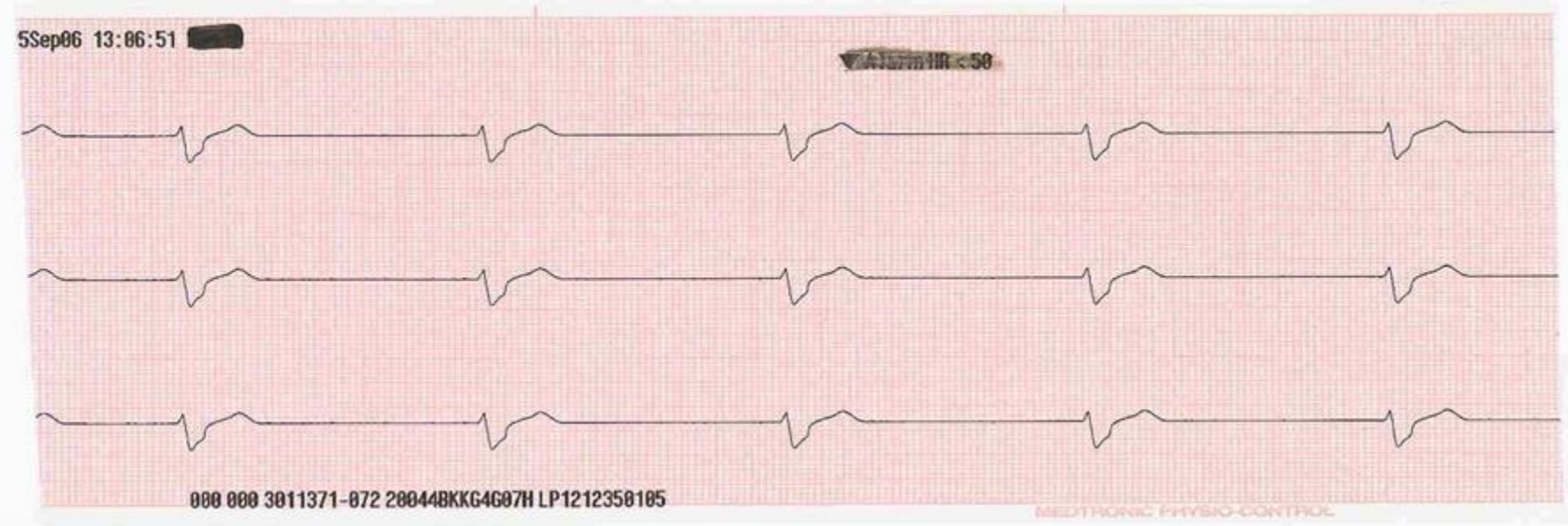
Electrical Activity? Pulse?

QRS Rate Width Regular / Regularly Irregular / Irregular

Atria P – Waves / Atrial Activity? Correlation

Lead II ECG

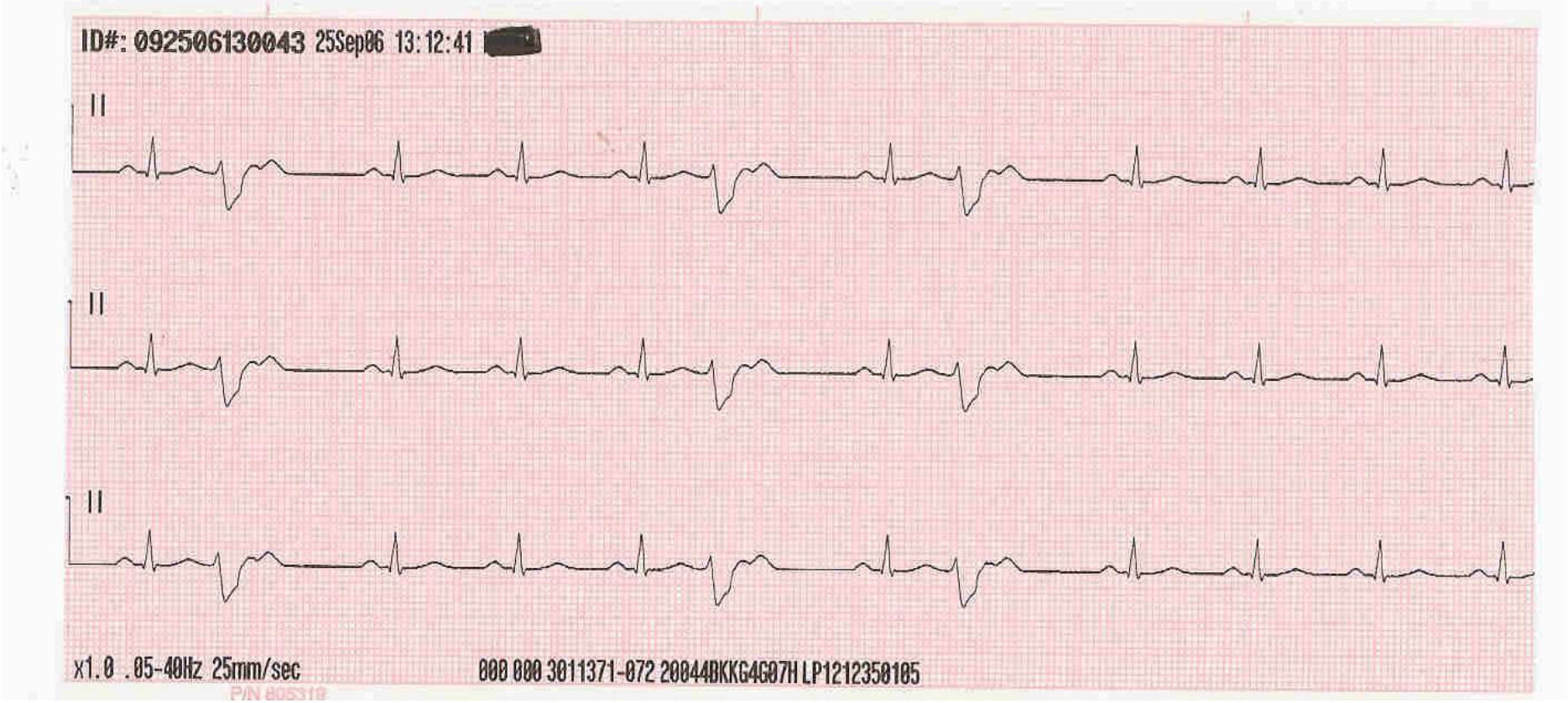
Pulseless Electrical Activity (PEA)



- **No Pulse** *with electrical activity*
- Often wide complexes

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Lead II ECG



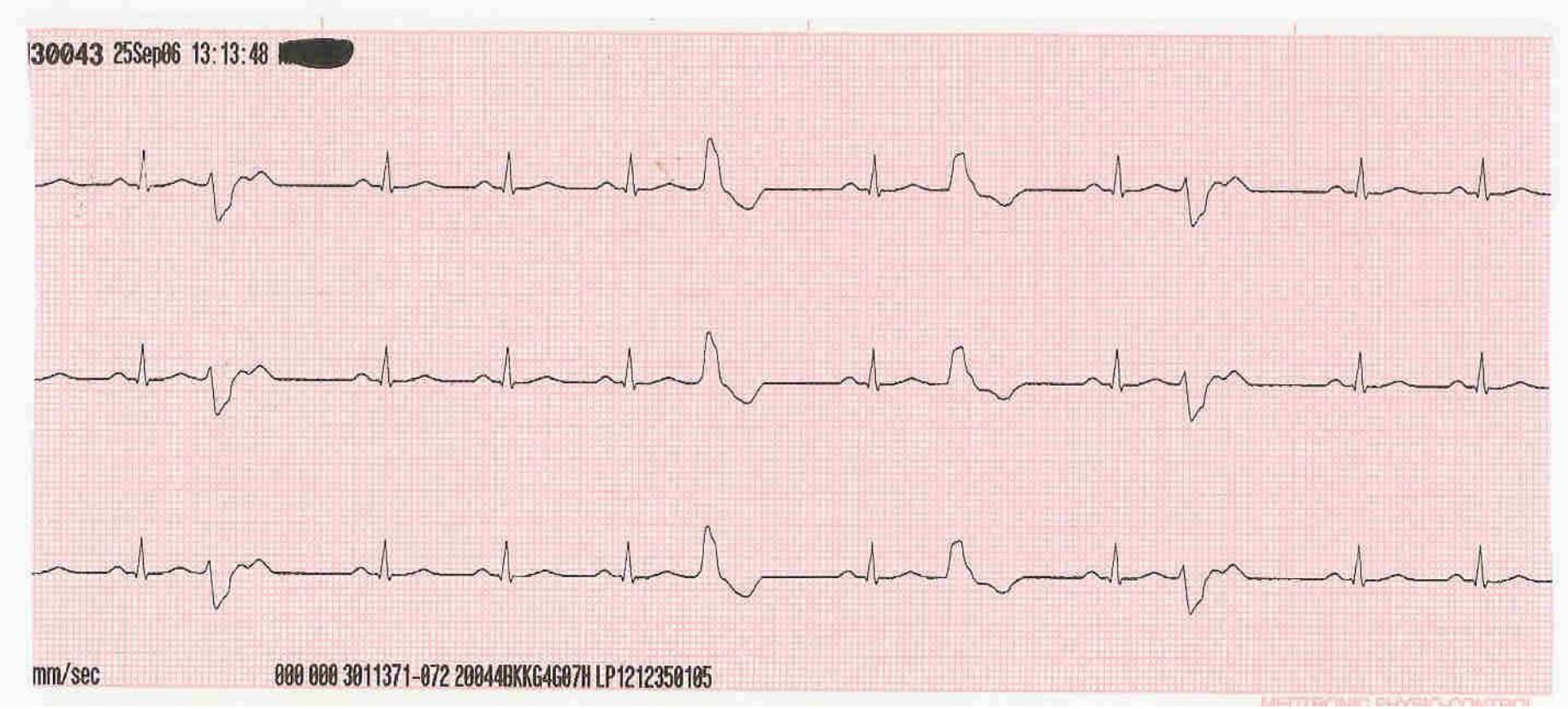
Rhythm

Electrical Activity? Pulse?

QRS Rate Width Regular / Regularly Irregular / Irregular

Atria P – Waves / Atrial Activity? Correlation

Lead II ECG



Rhythm

Electrical Activity? Pulse?

QRS Rate Width Regular / Regularly Irregular / Irregular

Atria P – Waves / Atrial Activity? Correlation

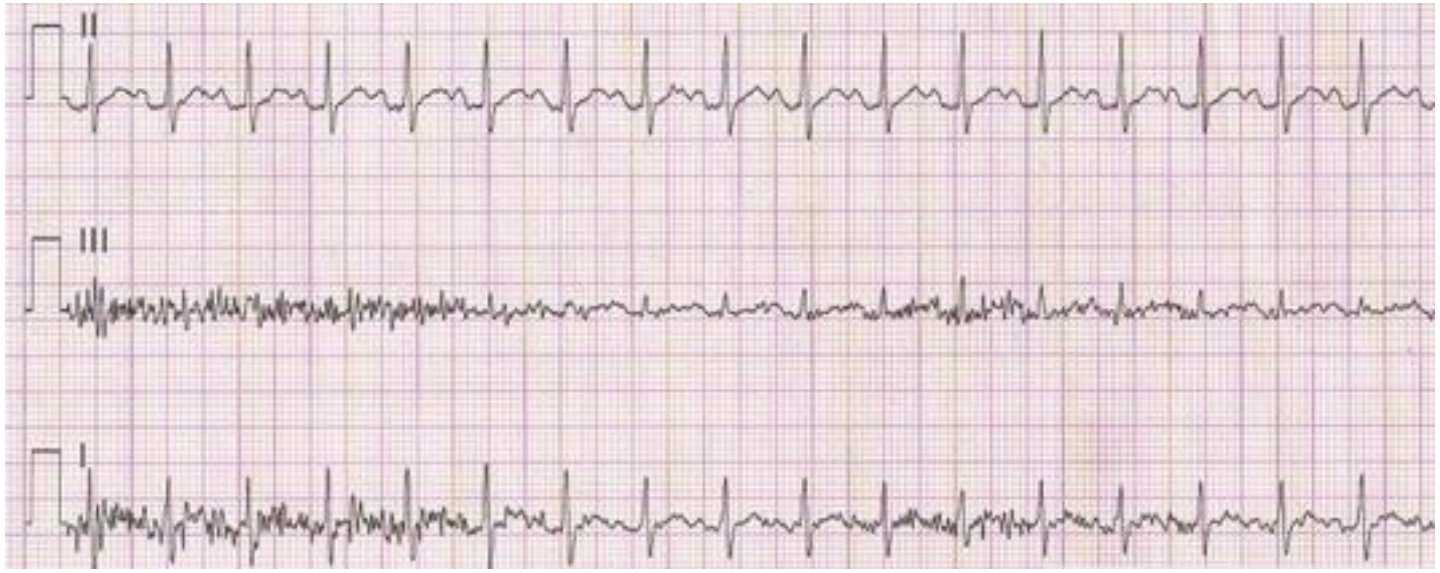
Lead II ECG

Artefact

Muscular or electrical artefact (interference) can disrupt the ECG.

Move electrode from large muscle mass

Turn off electrical equipment if necessary



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Lead II ECG

Summary

ECG in relation to the cardiac cycle

Lead placement

Basic checks

Structured analysis

Arrest rhythms

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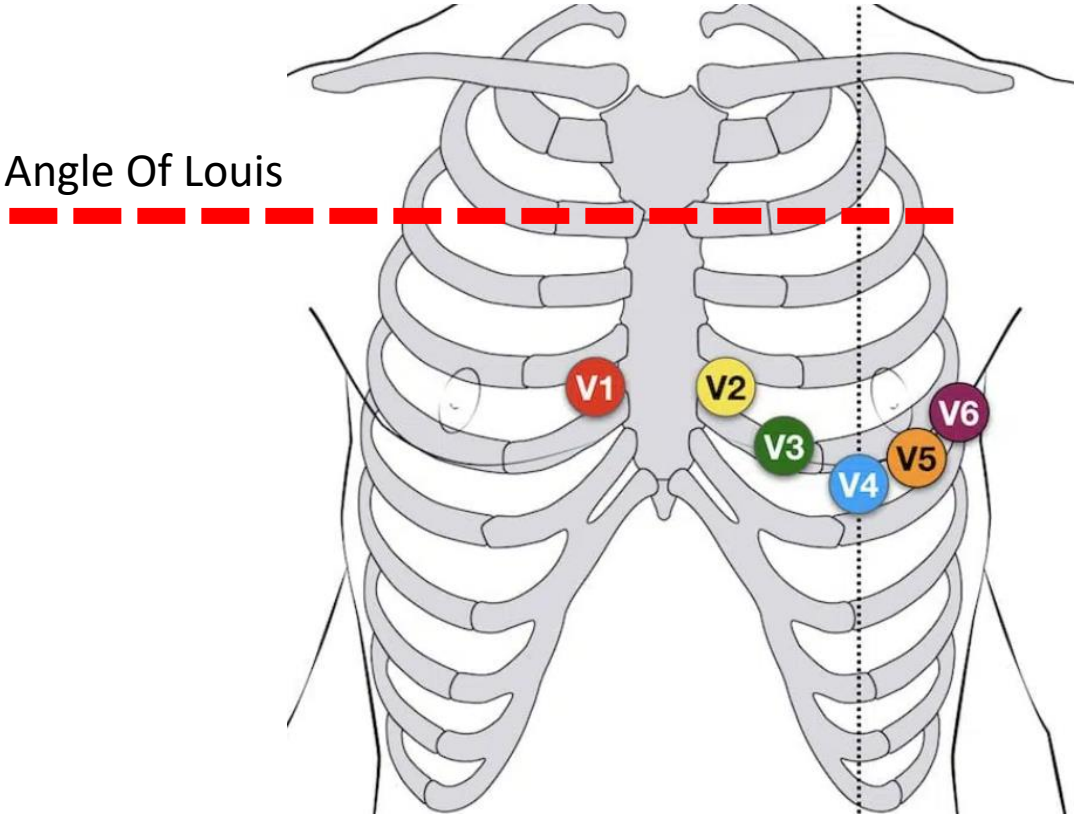
Lead II ECG

Any Questions



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12 lead ECG electrode placement



V1: 4th intercostal space (ICS), right sternal border

V2: 4th ICS along left sternal border

V4: 5th ICS, MCL

V3: Midway between V2 and V4

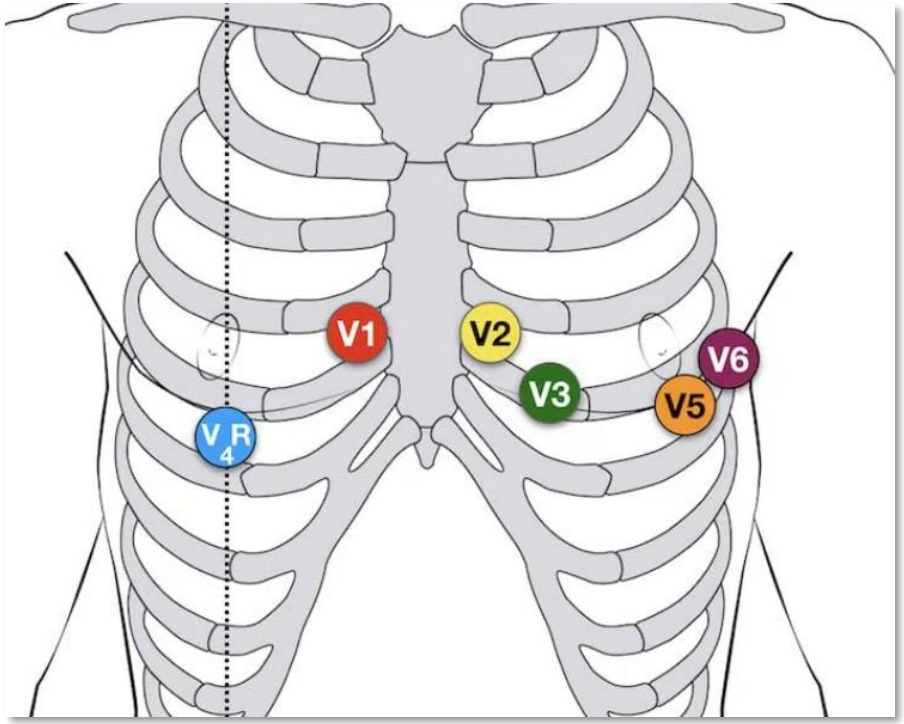
V5: 5th ICS, AAL(same level as V4)

V6: 5th ICS, MAL (same level as V4)

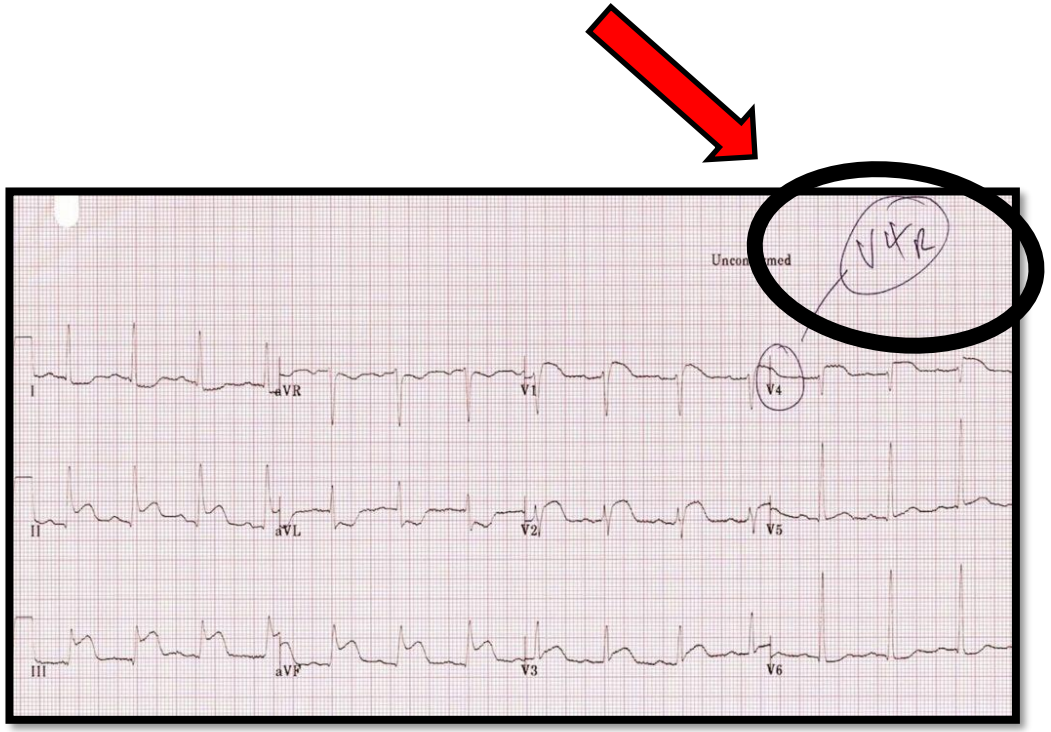
JRCALC: 10 rules and electrode placement

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12 lead ECG: lead placement for V4R trace

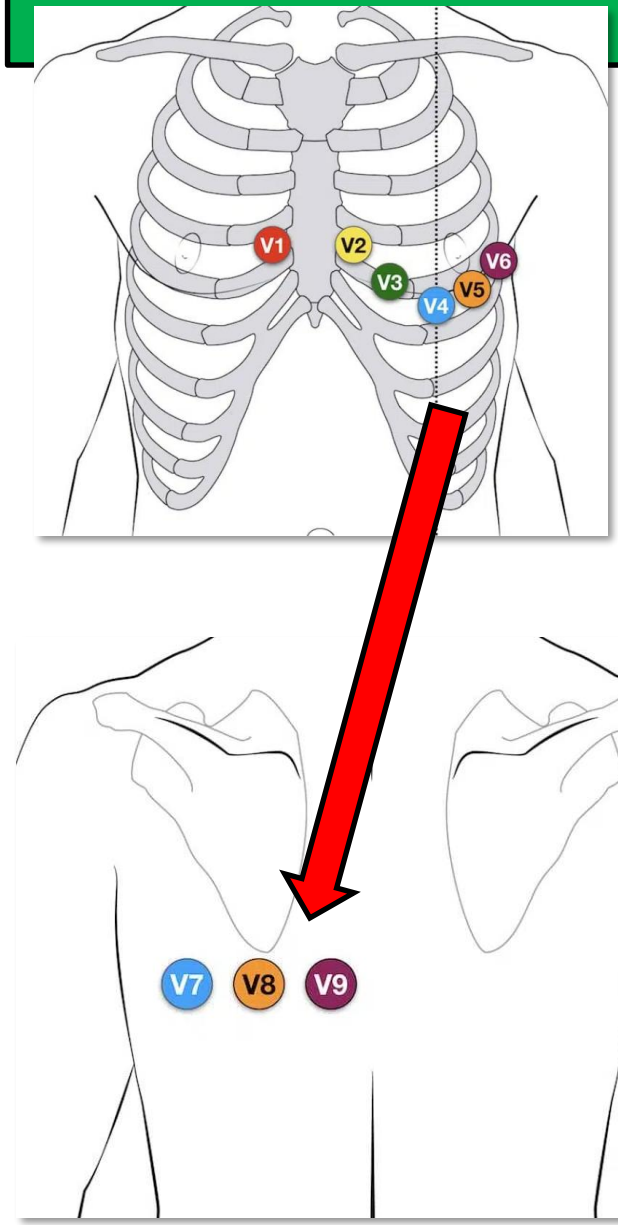


- Remove V4 from the chest.
- Put a new electrode on the lead
- Place the lead on 4th ICS, MCL on the left
- Reprint the 12 Lead
- Label the ECG “V4R”

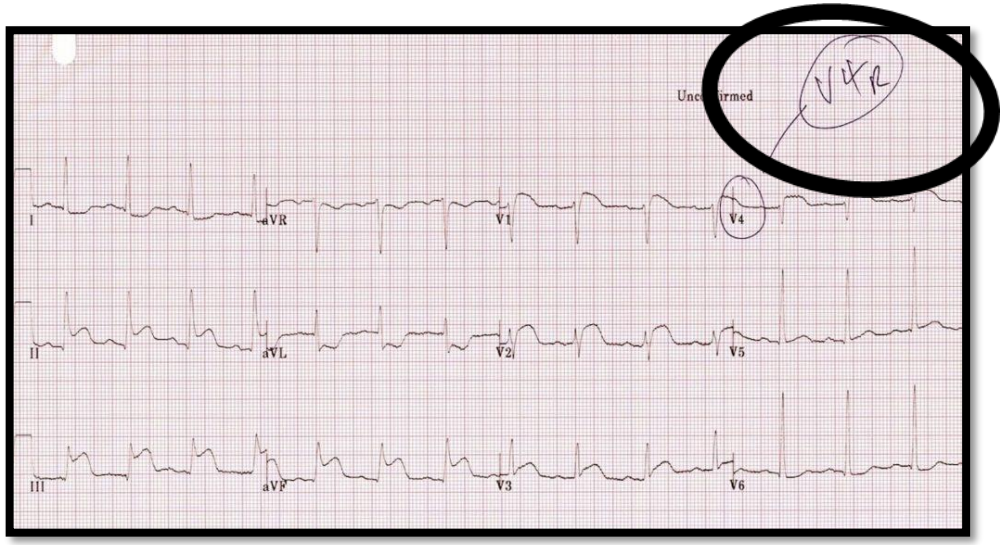


Link to JRCALC STEMI guidance:
Scroll down to RVI

12 lead ECG: Posterior ECG



- Remove V4, V5 and V6
- Put new electrodes on these leads
- Place them on the patient's back as shown
- These now become V7, V8 and V9
- Print off another 12 lead ECG annotating 'posterior ECG' at the top
- Cross out V4, V5 and V6 and re-label it 'V7, V8, V9'



JRCALC posterior MI

12 lead ECG

Any questions?

Further Reading

- JRCALC

- BMJ Learning
- (Free log-in)

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